GEOMETRIC REGULARITY DESCENDS FROM A DIRECTED COLIMIT OF SUBFIELDS TO THE FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite purely inseparable extensions are finitely presented. Used at the start of the counted proof (lines 47427-47429). Lets the finitely many generators and relations of the extension be moved into one of the subfields k_i.
- Change of base along a directed colimit of fields. Used at the end of the counted proof (lines 47431-47432). Identifies A tensor over k with k' with A tensor over k_i with k_i', so the hypothesis over k_i applies.

Lemma 0.1. *Let k be a field. Let A be an algebra over k. Let $k = \operatorname{colim} k_i$ be a directed colimit of subfields. If A is geometrically regular over each k_i, then A is geometrically regular over k.*

Proof. Let k'/k be a finite purely inseparable field extension. We can get k' by adjoining finitely many variables to k and imposing finitely many polynomial relations. Hence we see that there exists an i and a finite purely inseparable field extension k'_i/k_i such that $k_i = k \otimes_{k_i} k'_i$. Thus $A \otimes_k k' = A \otimes_{k_i} k'_i$ and the lemma is clear. □

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